

# PRADEEP S

📍 ECE (Major) & CSE (Minor) - CEG'27, Anna University, Chennai | CGPA: 7.99  
📞 +91 9342782799 | [pradeepsaravana01@gmail.com](mailto:pradeepsaravana01@gmail.com) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

## EXECUTIVE SUMMARY

ECE undergraduate with hands-on experience in Embedded C, bare-metal firmware development, and peripheral driver design across UART, SPI, I<sup>2</sup>C, timers, and interrupts. Experienced in microcontroller-based system design, motion-control firmware, and RTL design through academic projects and internships.

## TECHNICAL SKILLS

- **Programming:** C, Embedded C, C++, Python
- **Embedded Systems:** ARM Cortex-M4 (STM32F446RE), PIC16F877A, Bare-Metal Programming, Peripheral Driver Development, UART, SPI, I<sup>2</sup>C, Timers, Interrupts, FSM
- **Digital Design:** Verilog, RTL Design, RISC-V, AXI
- **Development Tools:** STM32CubeIDE, MPLAB X IDE, Vivado, ModelSim, EasyEDA, Git, Linux, VS Code

## PROJECTS

### Hybrid 3D Printer (SLA + FDM) ([View Here](#)) Dec 2024 – Jan 2025

- Led electronics integration including motor drivers, sensor interfacing, power distribution, and hardware debugging for a hybrid SLA + FDM 3D printer.
- Customized Marlin firmware and implemented G-code based control logic for synchronized resin extrusion, UV curing, and motion control.

### Secure Bootloader and OTA Firmware Update Framework ([View Here](#)) Jul 2026 – Present

- Developing a custom bootloader on STM32F446RE with packet-based UART communication and Flash memory management, supporting Flash erase and data programming.
- Extending the bootloader toward secure OTA firmware updates with firmware validation, application/bootloader separation, rollback protection, and integrity verification.

### True Random Number Generator (TRNG) – Tiny Tapeout (Sky130) ([View Here](#)) May 2026 – Present

- Designed a Verilog-based hardware TRNG using ring-oscillator jitter as an entropy source, with sampling and entropy-conditioning logic.
- Implementing entropy conditioning through sampling, Von Neumann correction, health testing, and output buffering, targeting Tiny Tapeout ASIC fabrication.

### RISC-V Data Memory AXI IP ([View Here](#)) Jan 2026

- Designed an RV32I data-memory block supporting load/store operations, byte/halfword access, address decoding, and sign/zero extension.
- Integrated the memory as an AXI-compliant IP and verified functionality through RTL simulation and waveform analysis using Vivado.

## EXPERIENCE

### Embedded Systems Intern – Emertxe Information Technologies ([View Here](#)) Dec 2025 – Jan 2026

- Developed Embedded C applications on PIC16F877A using timers, interrupts, bit manipulation, peripheral interfacing, and modular driver-based programming.
- Built a Microwave Oven Controller in PICSimLab featuring FSM-based control, keypad/LCD interfacing, multiple cooking modes, and timer management.

### Wireless Transceiver Design Intern — CWISD, Anna University Jun 2025

- Implemented ASK, PSK, QAM, and OFDM modulation using GNU Radio with signal generation, channel simulation, and performance evaluation.
- Designed digital filters and signal flow graphs while gaining exposure to baseband processing and wireless transceiver architecture.

## CERTIFICATIONS

- **Mastering Microcontroller and Embedded Driver Development**(FastBit Academy) ([View](#))
- **Letter of Appreciation** – Hybrid 3D Printer Contribution (CTF) ([View](#))
- **Fundamentals of Verilog** (Udemy) ([View](#))
- **Introduction to Internet of Things** (NPTEL) ([View](#))

## LANGUAGES

- English – Fluent | Tamil – Native | Japanese – Elementary